

2018 한국디지털포렌식학회 동계 학술대회

**물리메모리 덤프 파일조작을 통한
메모리 분석도구의 우회가능성 및 교차 검증의
필요성에 관한 연구**

2018.11.30

BoB

이한얼 김태우 전은진 한지혁 김한솔 김종민 김종현

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[2018년 5대 보안위협]



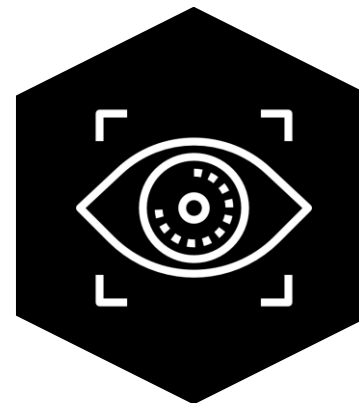
진화하는
공격자와 악성코드



날로 증가하는
제로데이 공격



진화하는
랜섬웨어



생체인증수단
우회공격 증가

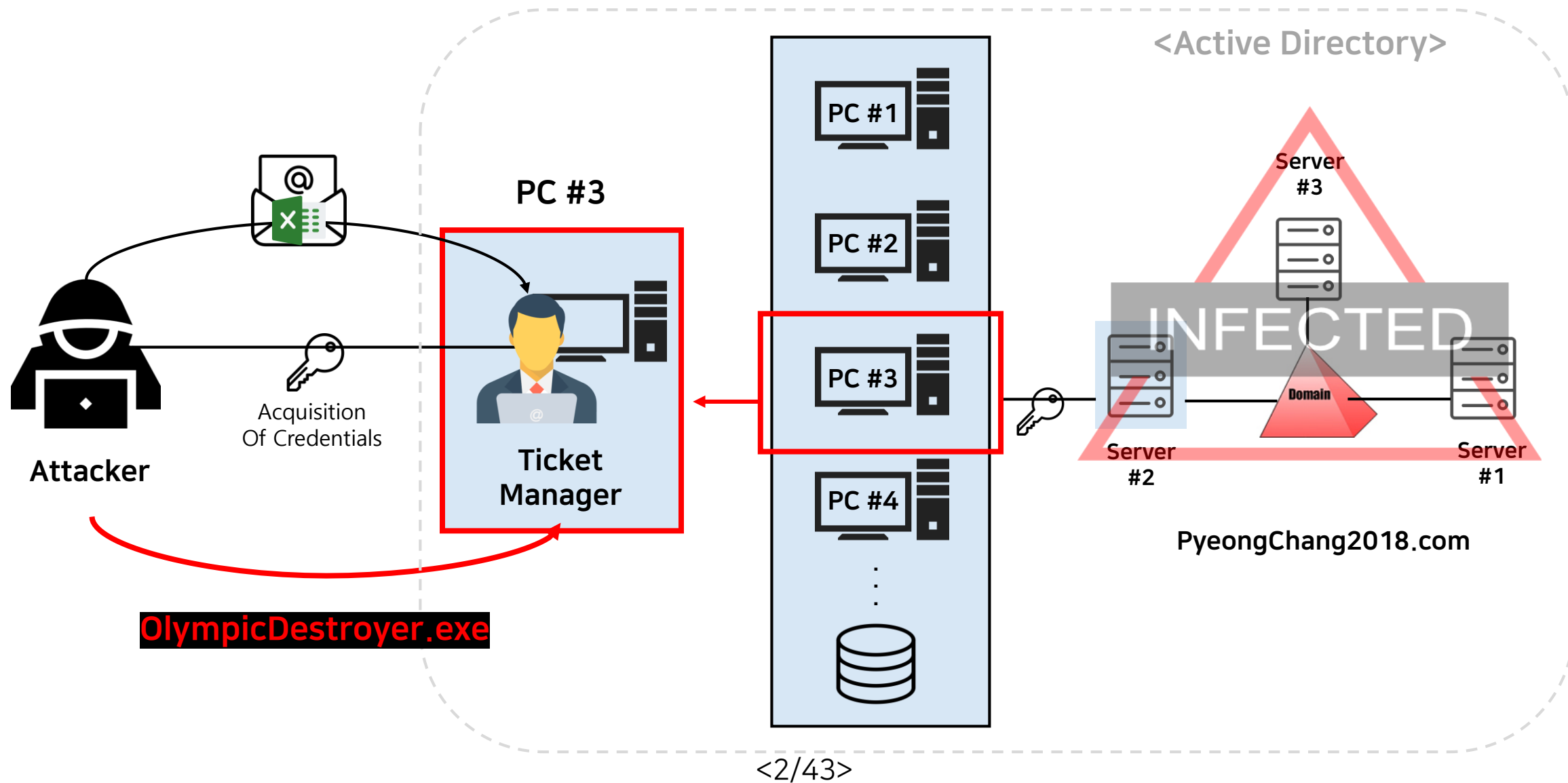


신뢰받는 SW
대상 공격 증가

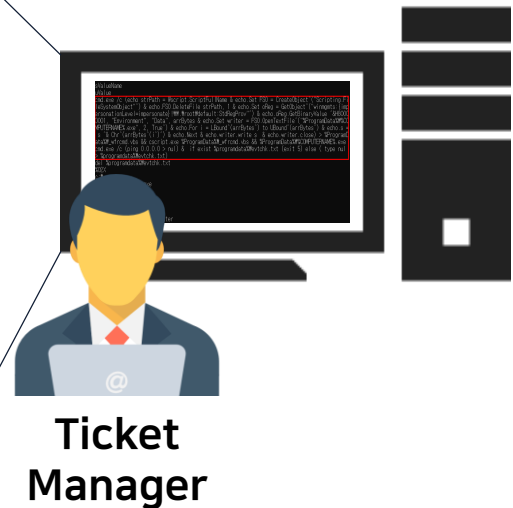
1

서론

연구 목적



```
sValueName  
uValue  
cmd.exe /c (echo strPath = Wscript.ScriptFullName & echo.Set FSO = CreateObject^("Scripting.Fi  
leSystemObject"^) & echo.FSO.DeleteFile strPath, 1 & echo.Set oReg = GetObject^("winmgmts:{imp  
ersonationLevel=impersonate}!###.root#default:StdRegProv"^) & echo.oReg.GetBinaryValue ^&H8000  
0001, "Environment", "Data", arrBytes & echo.Set writer = FSO.OpenTextFile^("%ProgramData%##CO  
MPUTERNAME%.exe", 2, True^) & echo.For i = LBound^ (arrBytes^) to UBound^ (arrBytes^) & echo.s =  
s ^& Chr^ (arrBytes^ (i)^) & echo.Next & echo.writer.write s & echo.writer.close) > %ProgramD  
ata%##_wfrcmd.vbs && cscript.exe %ProgramData%##_wfrcmd.vbs && %ProgramData%##_COMPUTERNAME%.exe  
cmd.exe /c (ping 0.0.0.0 > nul) & if exist %programdata%#evtchk.txt (exit 5) else ( type nul  
> %programdata%#evtchk.txt)  
del %programdata%#evtchk.txt  
%02X  
c:##  
#system32#notepad.exe  
BIN  
NBIN  
%s##root#directory#LDAP  
WQL  
SELECT ds_cn FROM ds_computer
```



1

서론

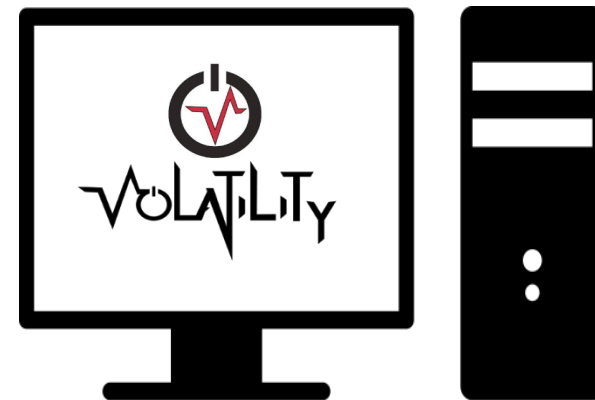
연구 목적



메모리
수집도구



덤프 파일



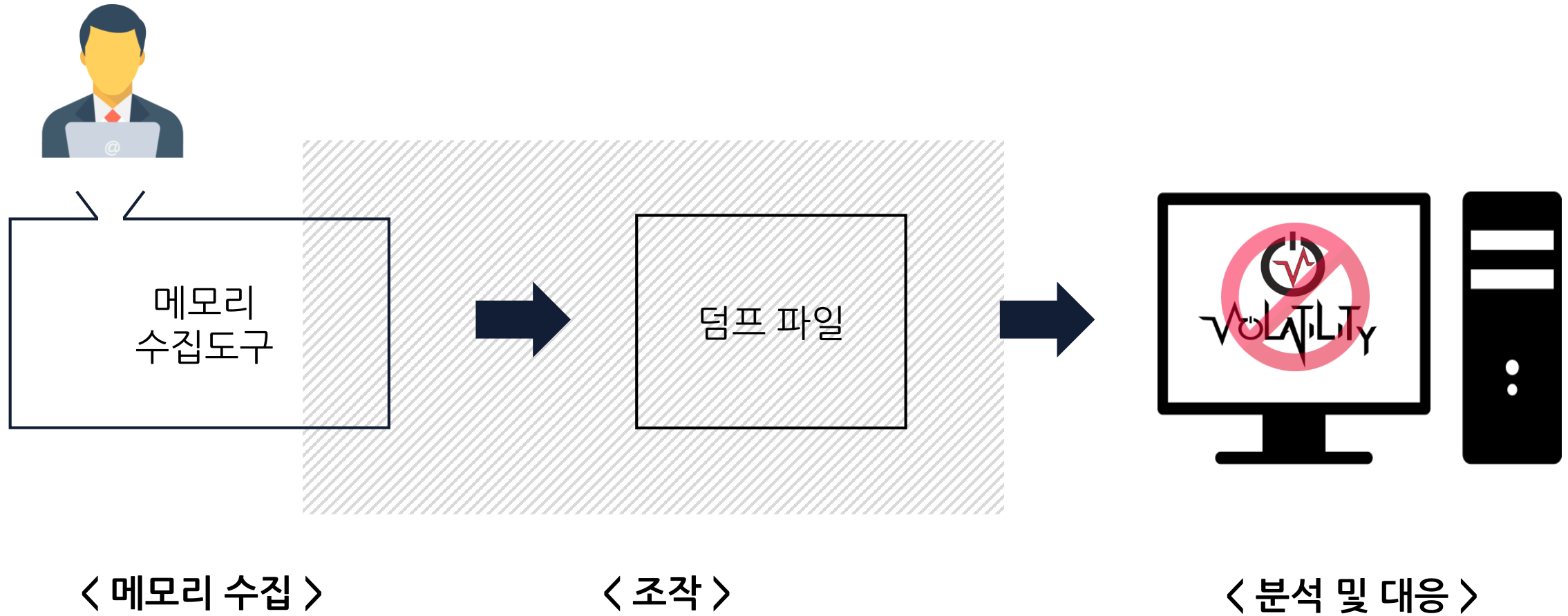
< 메모리 수집 >

< 분석 및 대응 >

1

서론

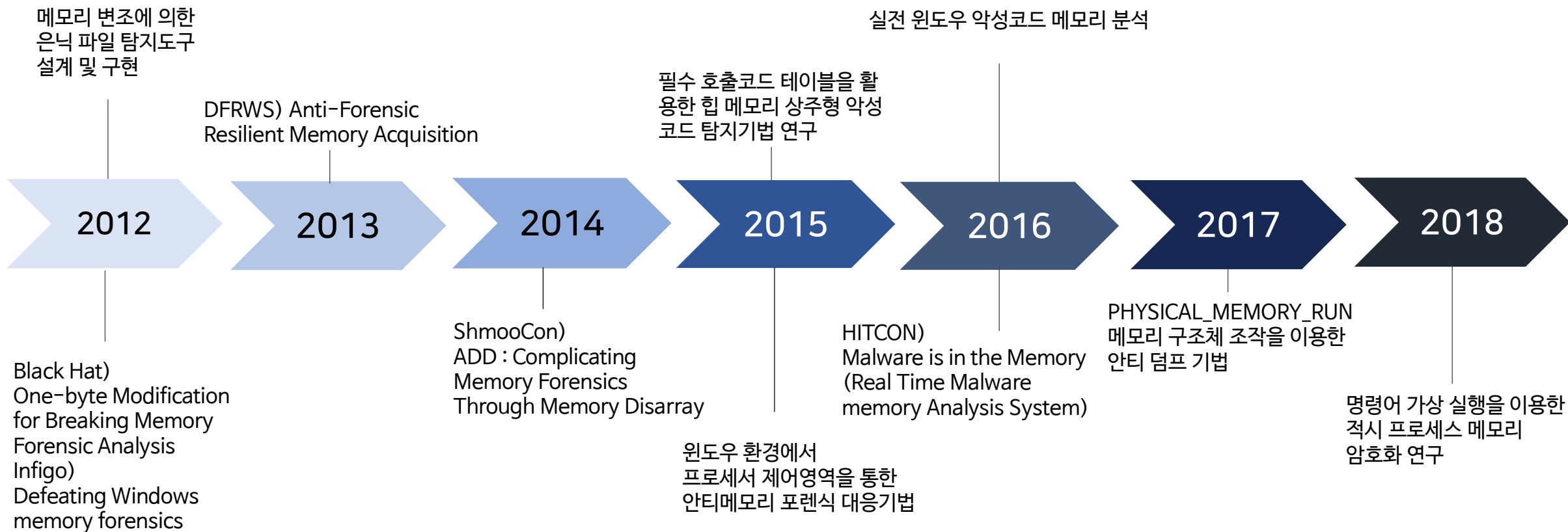
연구 목적



1

서론

관련 연구

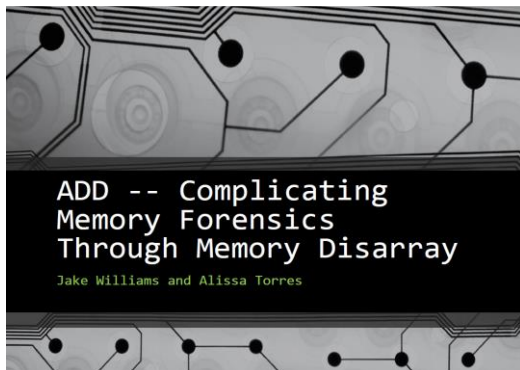


1

서론

관련 연구

메모리 변조에 의한
은닉 파일 탐지도구
설계 및 구현



DFRWS) Anti-Forensic
Resilient Memory Acquisition

필수 호출코드 테이블을 활
용한 힙 메모리 상주형 악성
코드 탐지기법 연구

2012

2013

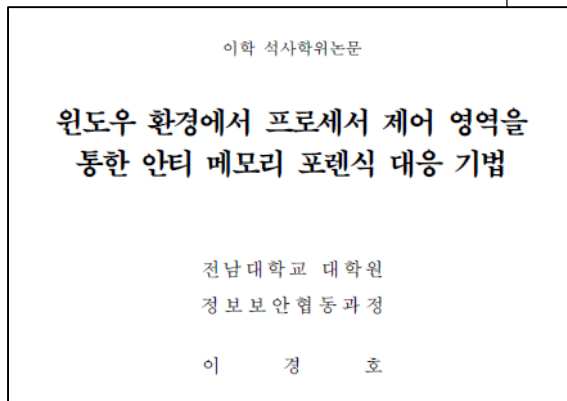
2014

2015

2016

2017

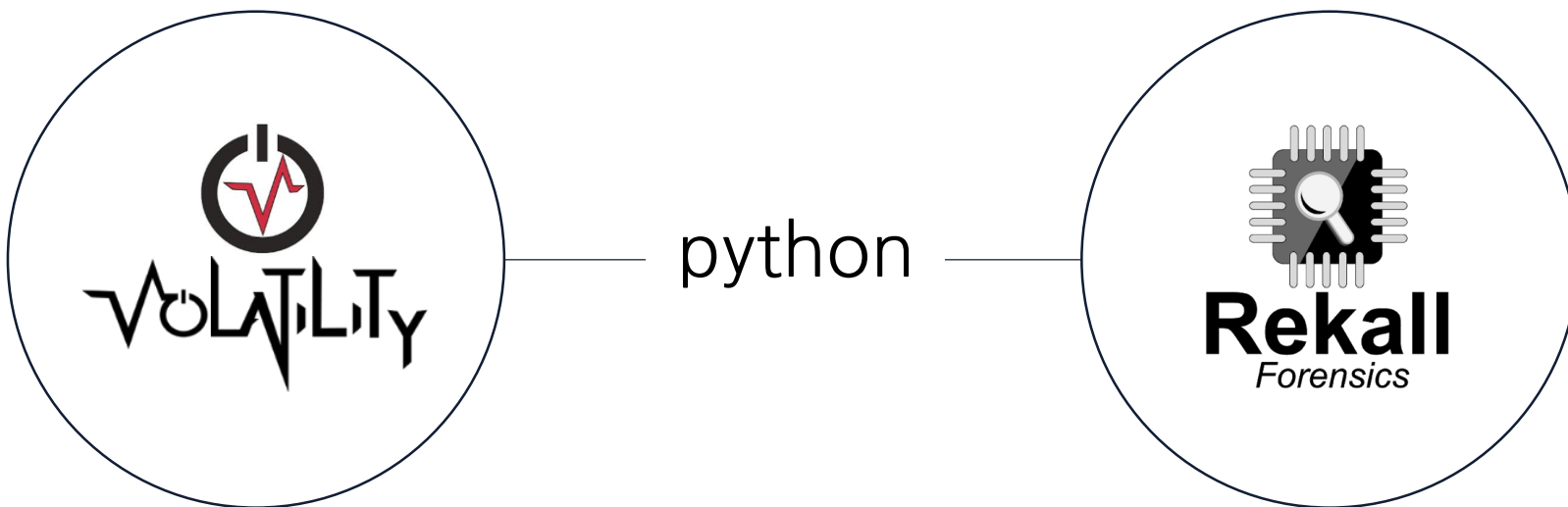
2018



HITCON)
Malware is in the Memory
(Real Time Malware
memory Analysis System)

PHYSICAL_MEMORY_RUN
메모리 구조체 조작을 이용한
안티 덤프 기법

명령어 가상 실행을 이용한
적시 프로세스 메모리
암호화 연구



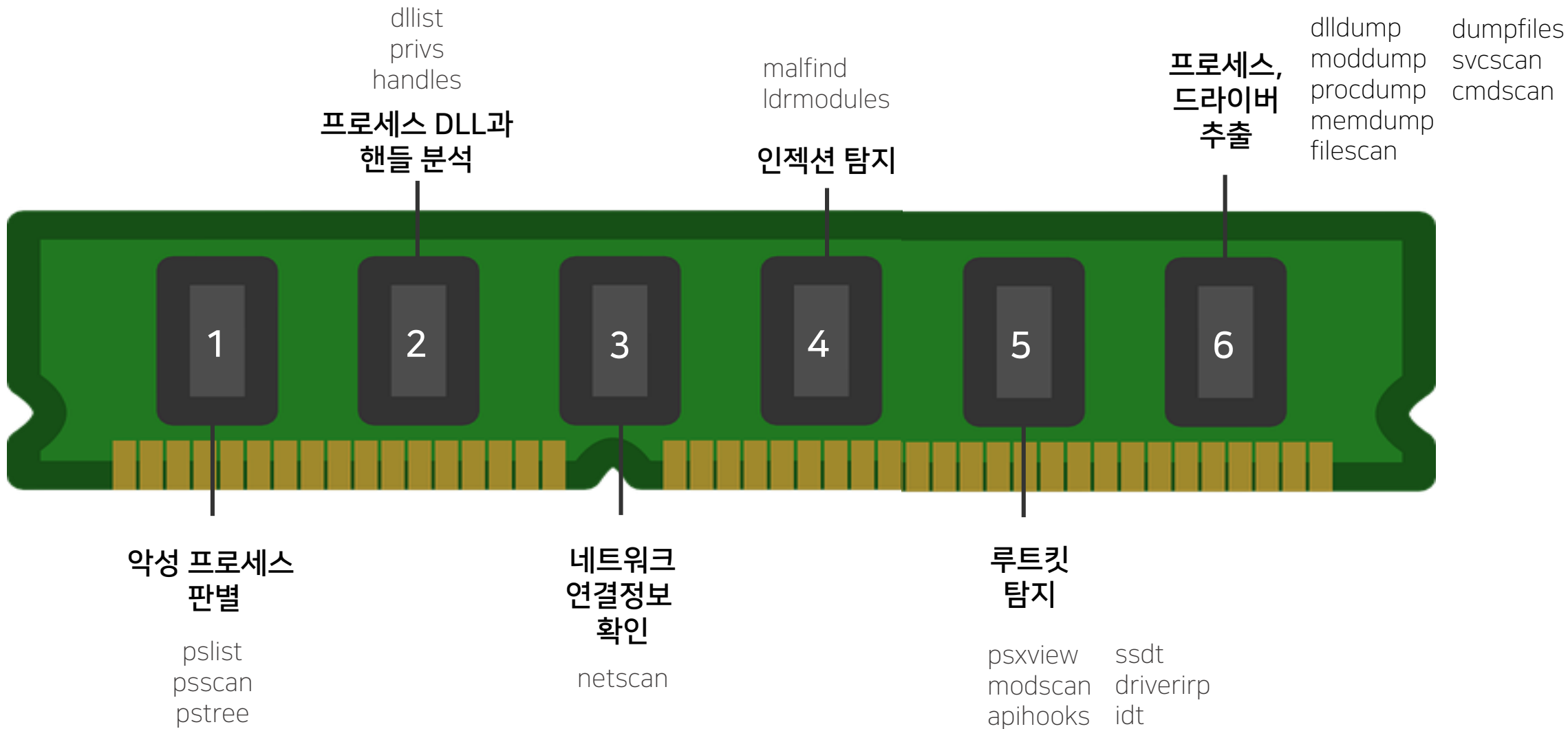
- Volatility Foundation 제작

- Google(GRR 프로젝트) 제작
- Volatility 기반
- System profile 자동 검색

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우회 방법

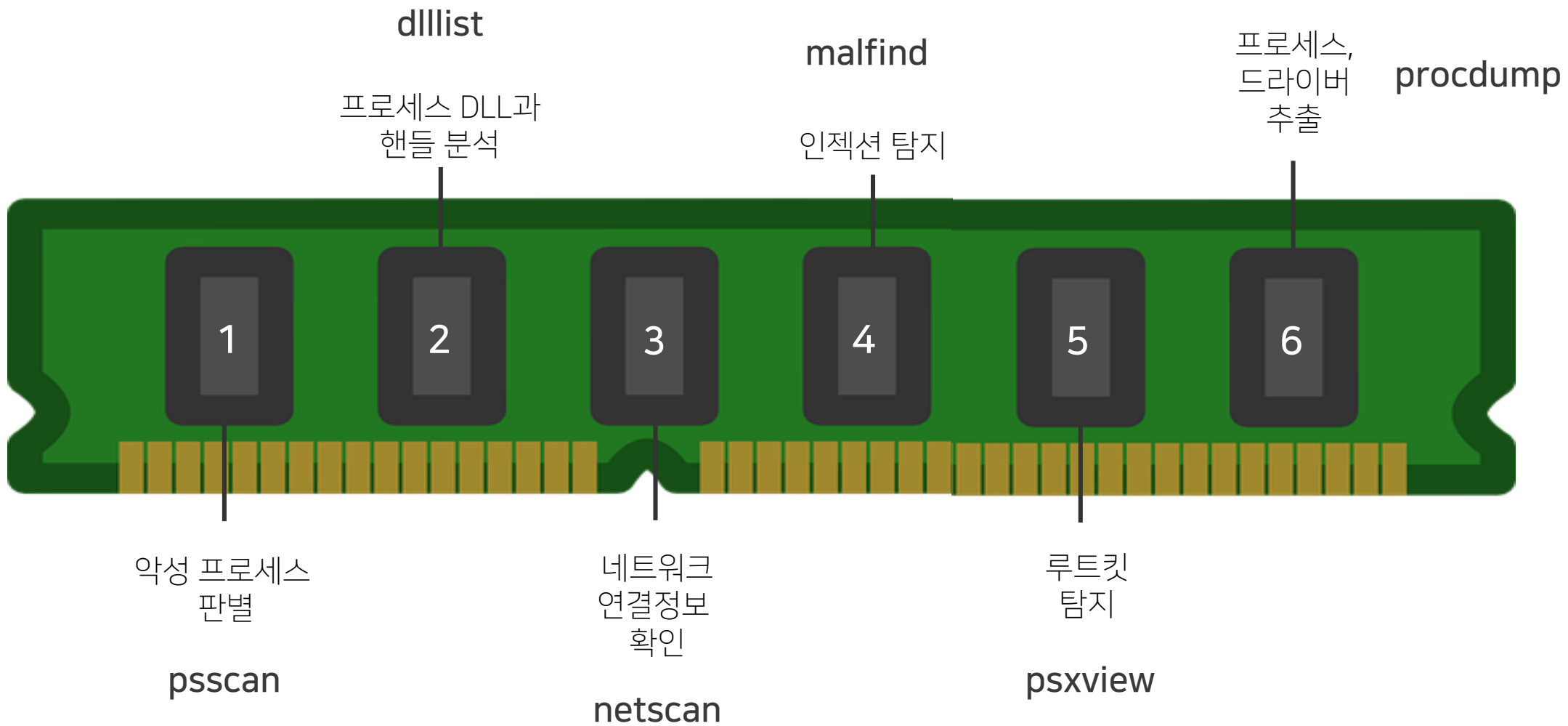
메모리 분석 절차



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우회 방법

메모리 분석 절차

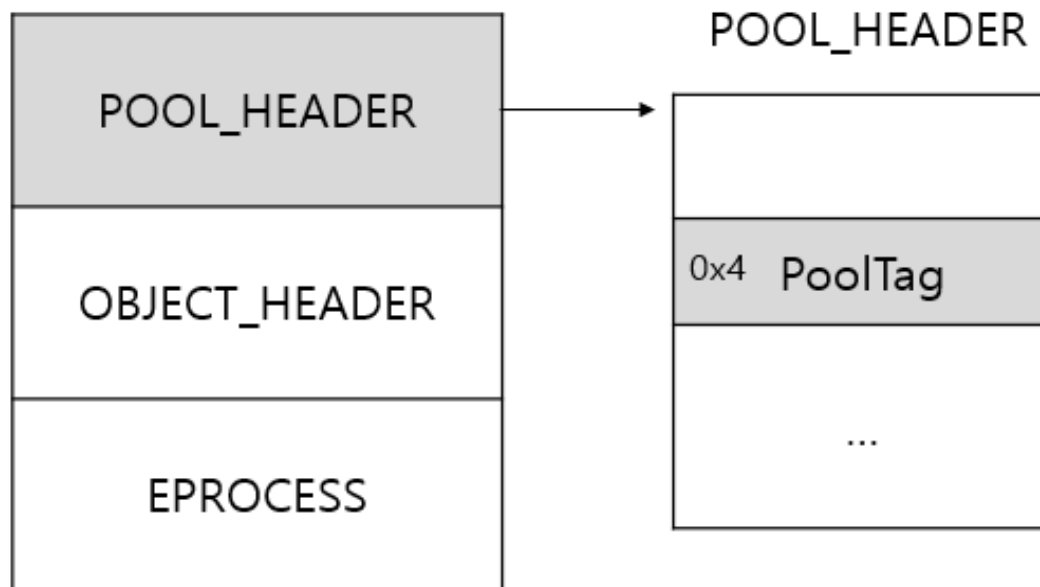


3

우회 방법

1) 프로세스 은닉

▶ psscan



3

우회 방법

1) 프로세스 은닉

▶ psscan

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psscan
Volatility Foundation Volatility Framework 2.6
Offset(P)          Name                PID   PPID PDB                Time created
-----
0x000000001d603b30 malware.exe          3180   656 0x000000000bbcf000 2018-10-11 12:23:59 UTC+0000
0x000000001d831280 wordpad.exe          3116   656 0x0000000015e06000 2018-10-11 12:23:46 UTC+0000
0x000000001d837b30 node.exe             3168  2856 0x00000000123c2000 2018-10-11 12:23:47 UTC+0000
0x000000001d863b30 SearchIndexer.       3288   496 0x000000001095f000 2018-10-11 12:23:47 UTC+0000
0x000000001d8b0630 wmpnetwk.exe         3488   496 0x000000001aa15000 2018-10-11 12:23:48 UTC+0000
0x000000001d8fab30 SearchFilterHo       3616  3288 0x000000000624a000 2018-10-11 12:23:49 UTC+0000
0x000000001d8fcb30 svchost.exe          3648   496 0x000000001498c000 2018-10-11 12:23:49 UTC+0000
```

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우회 방법

1) 프로세스 은닉

▶ psscan

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psscan
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	PPID	PDB	Time created
0x000000001d603b30	malware.exe	3180	656	0x000000000bbcf000	2018-10-11 12:23:59 UTC+0000
0x000000001d831280	wordpad.exe	3116	656	0x0000000015e06000	2018-10-11 12:23:46 UTC+0000
0x000000001d837b30	node.exe	3168	2856	0x00000000123c2000	2018-10-11 12:23:47 UTC+0000
0x000000001d863b30	SearchIndexer.	3288	496	0x000000001095f000	2018-10-11 12:23:47 UTC+0000
0x000000001d8b0630	wmpnetwk.exe	3488	496	0x000000001aa15000	2018-10-11 12:23:48 UTC+0000
0x000000001d8fab30	SearchFilterHo	3616	3288	0x000000000624a000	2018-10-11 12:23:49 UTC+0000
0x000000001d8fcb30	svchost.exe	3648	496	0x000000001498c000	2018-10-11 12:23:49 UTC+0000

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psscan
Volatility Foundation Volatility Framework 2.6
```

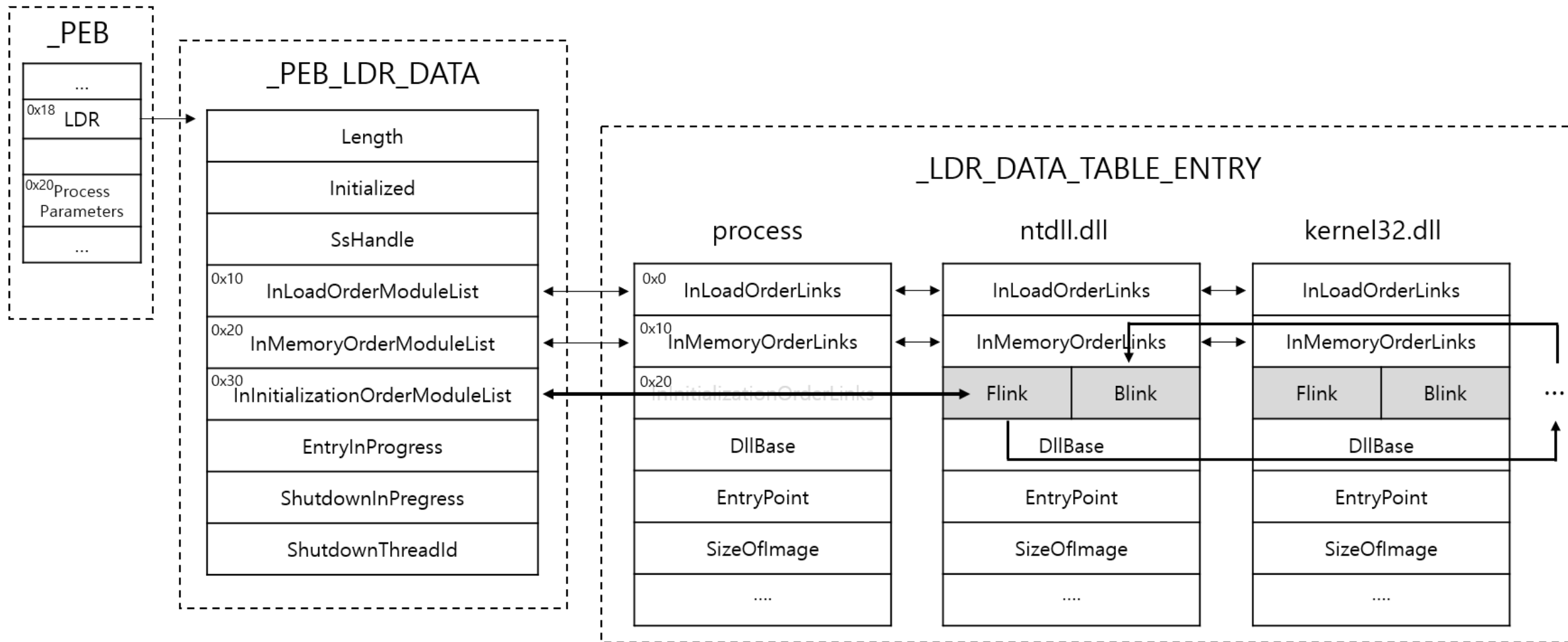
Offset(P)	Name	PID	PPID	PDB	Time created
0x000000001d831280	wordpad.exe	3116	656	0x0000000015e06000	2018-10-11 12:23:46 UTC+0000
0x000000001d837b30	node.exe	3168	2856	0x00000000123c2000	2018-10-11 12:23:47 UTC+0000
0x000000001d863b30	SearchIndexer.	3288	496	0x000000001095f000	2018-10-11 12:23:47 UTC+0000
0x000000001d8b0630	wmpnetwk.exe	3488	496	0x000000001aa15000	2018-10-11 12:23:48 UTC+0000
0x000000001d8fab30	SearchFilterHo	3616	3288	0x000000000624a000	2018-10-11 12:23:49 UTC+0000
0x000000001d8fcb30	svchost.exe	3648	496	0x000000001498c000	2018-10-11 12:23:49 UTC+0000

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우회 방법

2) DLL Injection 탐지 우회

▶ dlllist



3

우회 방법

2) DLL Injection 탐지 우회

▶ dlllist

```
C:\volatility-master>vol.py -f "C:\test.vmem" --profile=Win7SP0x64 dlllist -p 3180
Volatility Foundation Volatility Framework 2.6
*****
malware.exe pid: 3180
Command line : C:\Windows\system32\wbem\malware.exe
Service Pack 1
```

Base	Size	LoadCount	Path
0x00000000ff830000	0x5f000	0xffff	C:\Windows\system32\wbem\malware.exe
0x00000000772a0000	0x1a9000	0xffff	C:\Windows\SYSTEM32\ntdll.dll
0x0000000077180000	0x11f000	0xffff	C:\Windows\system32\kernel32.dll
0x000007fefd410000	0x6b000	0xffff	C:\Windows\system32\KERNELBASE.dll
0x000007feff330000	0xdb000	0xffff	C:\Windows\system32\ADVAPI32.dll
0x000007feff160000	0x9f000	0xffff	C:\Windows\system32\msvcrt.dll
0x000007fefedb0000	0x1f000	0xffff	C:\Windows\SYSTEM32\sechost.dll
0x000007fefd660000	0x12d000	0xffff	C:\Windows\system32\RPCRT4.dll
0x0000000077080000	0xfa000	0xffff	C:\Windows\system32\USER32.dll

```
C:\volatility-master>vol.py -f "C:\test.vmem" --profile=Win7SP0x64 dlllist -p 3180
Volatility Foundation Volatility Framework 2.6
*****
malware.exe pid: 3180
Command line : C:\Windows\system32\wbem\malware.exe
Service Pack 1
```

Base	Size	LoadCount	Path
0x00000000ff830000	0x5f000	0xffff	C:\Windows\system32\wbem\malware.exe
0x0000000077180000	0x11f000	0xffff	C:\Windows\system32\kernel32.dll
0x000007fefd410000	0x6b000	0xffff	C:\Windows\system32\KERNELBASE.dll
0x000007feff330000	0xdb000	0xffff	C:\Windows\system32\ADVAPI32.dll
0x000007feff160000	0x9f000	0xffff	C:\Windows\system32\msvcrt.dll
0x000007fefedb0000	0x1f000	0xffff	C:\Windows\SYSTEM32\sechost.dll
0x000007fefd660000	0x12d000	0xffff	C:\Windows\system32\RPCRT4.dll
0x0000000077080000	0xfa000	0xffff	C:\Windows\system32\USER32.dll

3

우회 방법

3) 네트워크 연결정보 은닉

▶ netscan

STATE	PoolTag
TCP_LISTENER	TcpL(0x5463704C
TCP_ENDPOINT	TcpE(0x54637045)
UDP_ENDPOINT	UdpA(0x55647041)

```

1DB796D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
1DB796E0 12 00 32 02 54 63 70 45 00 00 00 00 00 00 00 00 ..2 TcpE.....
1DB796F0 10 40 3A 09 80 FA FF FF 00 00 00 00 00 00 00 00 .@:.éúÿÿ.....
1DB79700 40 9E B8 07 80 FA FF FF 10 33 24 07 80 FA FF FF @ž,.éúÿÿ.3$.éúÿÿ
1DB79710 80 B3 B8 08 80 FA FF FF 80 CC 1E 07 80 FA FF FF €³,.éúÿÿ€Ì..éúÿÿ
1DB79720 80 CC 1E 07 80 FA FF FF 00 92 C9 E1 00 00 00 00 €Ì..éúÿÿ.'Éá....
1DB79730 80 5C 2C 07 80 FA FF FF 80 5C 2C 07 80 FA FF FF €\,.éúÿÿ€\,.éúÿÿ

```

PoolTag : 숨기려고 하는 필드의 offset 주소에서 0xc 오프셋만큼 위에 위치

<16/43>

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우회 방법

3) 네트워크 연결정보 은닉

▶ netscan

0x1dee2a80	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System
0x1dee2a80	TCPv6	:::445	:::0	LISTENING	4	System
0x1d98d400	TCPv4	*:0	56.59.147.8:0	CLOSED	4	System
0x1d98dbe0	TCPv4	127.0.0.1:49159	127.0.0.1:5357	CLOSED	396	svchost.exe
0x1d9a5010	TCPv6	*:0	383b:9308:80fa:ffff:383b:9308:80fa:ffff:0	CLOSED	4	
0x1db796f0	TCPv4	192.168.79.133:49160	119.207.64.137:80	CLOSED	916	svchost.exe
0x1e029580	UDPv4	0.0.0.0:5355	*:*		916	svchost.exe
0x1e029580	UDPv6	:::5355	*:*		916	svchost.exe
0x1e06e450	UDPv4	0.0.0.0:51905	*:*		3648	svchost.exe
0x1e0b57d0	UDPv4	0.0.0.0:57390	*:*		916	svchost.exe
0x1e0b57d0	UDPv6	:::57390	*:*		916	svchost.exe

3

우회 방법

3) 네트워크 연결정보 은닉

▶ netscan

0x1dee2a80	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System
0x1dee2a80	TCPv6	:::445	:::0	LISTENING	4	System
0x1d98d400	TCPv4	-:0	56.59.147.8:0	CLOSED	4	System
0x1d98dbe0	TCPv4	127.0.0.1:49159	127.0.0.1:5357	CLOSED	396	svchost.exe
0x1d9a5010	TCPv6	-:0	383b:9308:80fa:ffff:383b:9308:80fa:ffff:0	CLOSED	4	
0x1db79cf0	TCPv4	192.168.79.133:49160	119.207.64.137:80	CLOSED	916	svchost.exe
0x1e029580	UDPv4	0.0.0.0:5355	*:*		916	svchost.exe
0x1e029580	UDPv6	:::5355	*:*		916	svchost.exe
0x1e06e450	UDPv4	0.0.0.0:51905	*:*		3648	svchost.exe
0x1e0b57d0	UDPv4	0.0.0.0:57390	*:*		916	svchost.exe
0x1e0b57d0	UDPv6	:::57390	*:*		916	svchost.exe

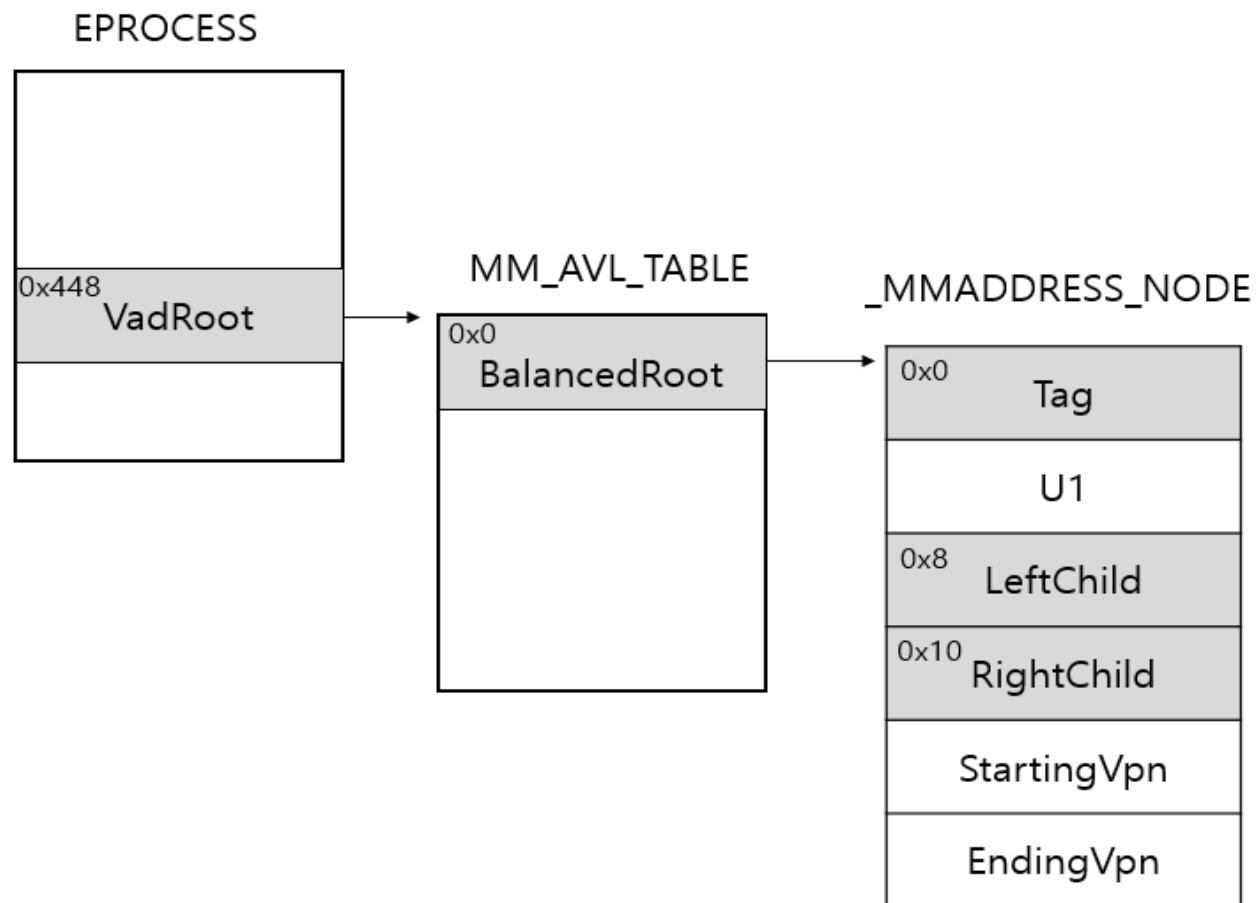
0x1dee2a80	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System
0x1dee2a80	TCPv6	:::445	:::0	LISTENING	4	System
0x1d98d400	TCPv4	-:0	56.59.147.8:0	CLOSED	4	System
0x1d98dbe0	TCPv4	127.0.0.1:49159	127.0.0.1:5357	CLOSED	396	svchost.exe
0x1d9a5010	TCPv6	-:0	383b:9308:80fa:ffff:383b:9308:80fa:ffff:0	CLOSED	4	
0x1e029580	UDPv4	0.0.0.0:5355	*:*		916	svchost.exe
0x1e029580	UDPv6	:::5355	*:*		916	svchost.exe
0x1e06e450	UDPv4	0.0.0.0:51905	*:*		3648	svchost.exe
0x1e0b57d0	UDPv4	0.0.0.0:57390	*:*		916	svchost.exe
0x1e0b57d0	UDPv6	:::57390	*:*		916	svchost.exe

3

우회 방법

4) Code Injection 탐지 우회

▶ malfind



3

우회 방법

4) Code Injection 탐지 우회

▶ malfind

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 malfind -p 3180
Volatility Foundation Volatility Framework 2.6
Process: malware.exe Pid: 3180 Address: 0x2560000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 2, PrivateMemory: 1, Protection: 6

0x02560000  00 00 00 00 00 00 00 00 68 a7 dc 9b c4 5f 00 01  .....h....._...
0x02560010  ee ff ee ff 00 00 00 00 28 01 56 02 00 00 00 00  .....(.V.....
0x02560020  28 01 56 02 00 00 00 00 00 00 56 02 00 00 00 00  (.V.....V.....
0x02560030  00 00 56 02 00 00 00 00 80 00 00 00 00 00 00 00  ..V.....
```

3

우회 방법

4) Code Injection 탐지 우회

▶ malfind

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 malfind -p 3180
Volatility Foundation Volatility Framework 2.6
Process: malware.exe Pid: 3180 Address: 0x2560000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 2, PrivateMemory: 1, Protection: 6

0x02560000  00 00 00 00 00 00 00 00 68 a7 dc 9b c4 5f 00 01  .....h....._...
0x02560010  ee ff ee ff 00 00 00 00 28 01 56 02 00 00 00 00  .....(V.....
0x02560020  28 01 56 02 00 00 00 00 00 00 56 02 00 00 00 00  (.V.....V.....
0x02560030  00 00 56 02 00 00 00 00 80 00 00 00 00 00 00 00  ..V.....
```

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 malfind -p 3180
Volatility Foundation Volatility Framework 2.6

C:\#volatility-master>
```

▶ psxview

7개의 필드 우회 -> 해당 프로세스 은닉

pslist

psscan

thrdproc

pspcid

csrss

session

deskthrd

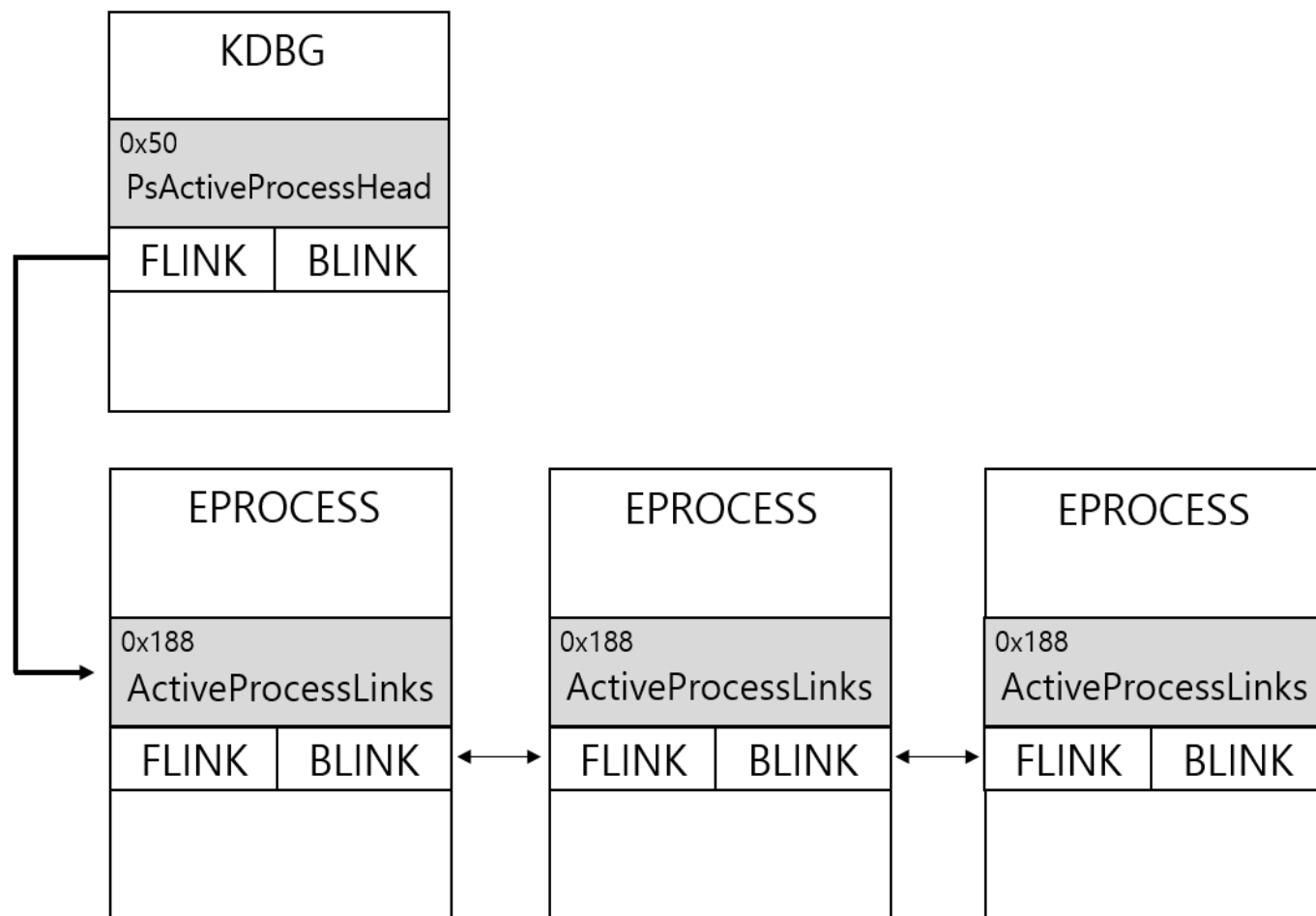
3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ pslist



3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ pslist

```
C:\volatility-master>vol.py -f C:\test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001de37580	VGAAuthService.	1688	True	True	True	True	True	True	True	
0x000000001e5f7b30	lsass.exe	512	True	True	True	True	True	True	False	
0x000000001da1bb30	vmtoolsd.exe	2644	True	True	True	True	True	True	True	
0x000000001e2c7060	svchost.exe	764	True	True	True	True	True	True	True	
0x000000001da11b30	explorer.exe	2336	True	True	True	True	True	True	True	
0x000000001e584b30	userinit.exe	2268	True	True	True	True	True	True	True	
0x000000001e302710	svchost.exe	860	True	True	True	True	True	True	True	
0x000000001de25b30	svchost.exe	1656	True	True	True	True	True	True	True	
0x000000001e32b9e0	svchost.exe	896	True	True	True	True	True	True	True	
0x000000001d603b30	malware.exe	3180	False	True	True	True	True	True	True	
0x000000001d831280	wordpad.exe	3116	True	True	True	True	True	True	True	
0x000000001df63a10	dwm.exe	2292	True	True	True	True	True	True	True	
0x000000001e078220	spoolsv.exe	1204	True	True	True	True	True	True	True	

3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ psscan

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001de37580	VGAAuthService.	1688	True	True	True	True	True	True	True	
0x000000001e5f7b30	lsass.exe	512	True	True	True	True	True	True	False	
0x000000001da1bb30	vmtoolsd.exe	2644	True	True	True	True	True	True	True	
0x000000001e2c7060	svchost.exe	764	True	True	True	True	True	True	True	
0x000000001da11b30	explorer.exe	2336	True	True	True	True	True	True	True	
0x000000001e584b30	userinit.exe	2268	True	True	True	True	True	True	True	
0x000000001e302710	svchost.exe	860	True	True	True	True	True	True	True	
0x000000001de25b30	svchost.exe	1656	True	True	True	True	True	True	True	
0x000000001e32b9e0	svchost.exe	896	True	True	True	True	True	True	True	
0x000000001d603b30	malware.exe	3180	True	False	True	True	True	True	True	
0x000000001d831280	wordpad.exe	3116	True	True	True	True	True	True	True	
0x000000001df63a10	dwm.exe	2292	True	True	True	True	True	True	True	
0x000000001e078220	spoolsv.exe	1204	True	True	True	True	True	True	True	

3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ thrddproc

EPROCESS

0x170	ExitTime
0x1e0	SessionProcessLinks
0x258	Win32Process

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)      Name                      PID pslist psscan thrddproc pspcid csrss session deskthrd ExitTime
-----
0x000000001de37580 VGAAuthService.           1688 True  True  True  True  True  True  True
0x000000001e5f7b30 lsass.exe                  512 True  True  True  True  True  True  True
0x000000001da1bb30 vmtoolsd.exe              2644 True  True  True  True  True  True  True
0x000000001e2c7060 svchost.exe                764 True  True  True  True  True  True  True
0x000000001da11b30 explorer.exe              2336 True  True  True  True  True  True  True
0x000000001e584b30 userinit.exe              2268 True  True  True  True  True  True  True
0x000000001e302710 svchost.exe                860 True  True  True  True  True  True  True
0x000000001de25b30 svchost.exe              1656 True  True  True  True  True  True  True
0x000000001e32b9e0 svchost.exe                896 True  True  True  True  True  True  True
0x000000001d603b30 malware.exe               3180 True  True  False True  True  True  True  2018-10-11 12:23:59 UTC+0000
0x000000001d831280 wordpad.exe                3116 True  True  True  True  True  True  True
0x000000001df63a10 dwm.exe                   2292 True  True  True  True  True  True  True
0x000000001e078220 spoolsv.exe               1204 True  True  True  True  True  True  True
```

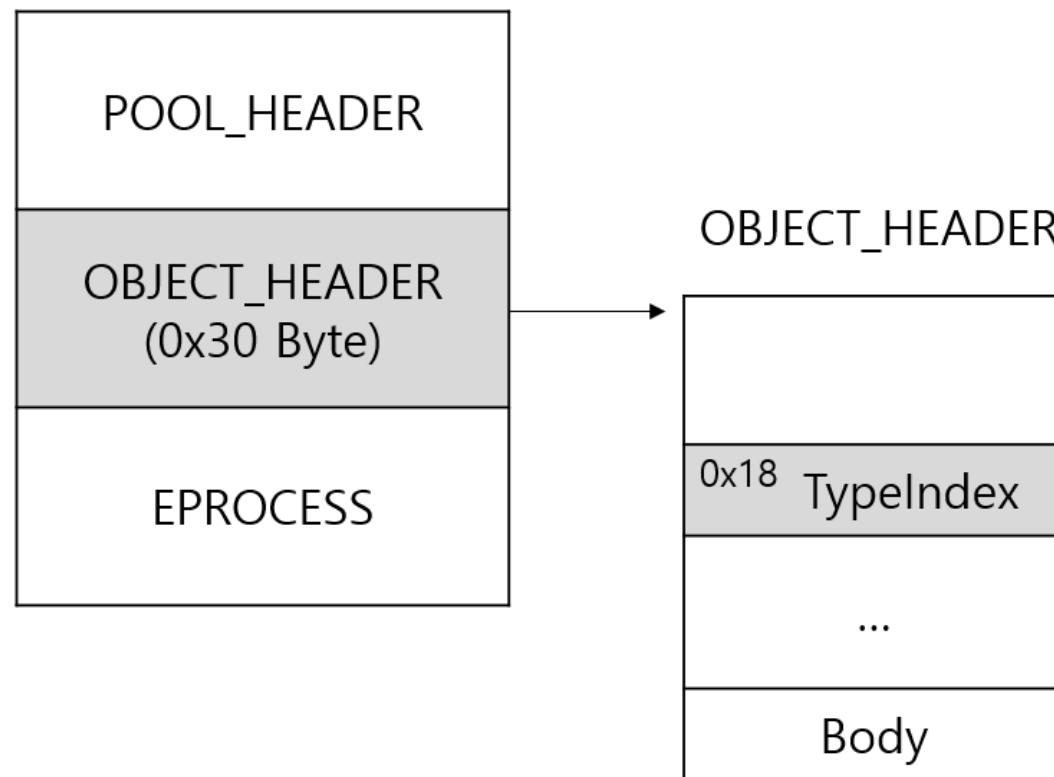
3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ pspcid, csrss



3

우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ pspcid, csrss

```
C:\#volatility-master>vol.py -f C:\test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001e38e9b0	svchost.exe	396	True	True	True	True	True	True	True	
0x000000001d8b0630	wmpnetwk.exe	3488	True	True	True	True	True	True	True	
0x000000001db1bb30	REPORTXSvc.exe	2876	True	True	True	True	True	True	True	
0x000000001e3dfb30	svchost.exe	916	True	True	True	True	True	True	True	
0x000000001dabb600	IMG5F50Start_x	2752	True	True	True	True	True	True	True	
0x000000001d9d7b30	calc.exe	3216	True	True	True	True	True	True	True	
0x000000001e333b30	svchost.exe	920	True	True	True	True	True	True	True	
0x000000001e6e9b30	csrss.exe	356	True	True	True	True	False	True	True	
0x000000001ef10b30	smss.exe	268	True	True	True	True	False	False	False	
0x000000001db6f9c0	cmd.exe	2544	True	True	False	True	False	True	False	2018-10-11 12:24:00 UTC+0000
0x000000001e561060	csrss.exe	444	True	True	True	True	False	True	True	
0x000000001ffbdb30	System	4	True	True	True	True	False	False	False	
0x000000001d603b30	malware.exe	3180	True	True	True	False	False	True	True	

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우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ session

EPROCESS

0x170	ExitTime
0x1e0	SessionProcessLinks
0x258	Win32Process

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)      Name                PID  pslist  psscan  thrdproc  pspcid  csrss  session  deskthrd  ExitTime
-----
0x000000001de37580  VGAuthService.      1688  True    True    True      True    True    True    True      True
0x000000001e5f7b30  lsass.exe           512   True    True    True      True    True    True    False     True
0x000000001da1bb30  vmtoolsd.exe        2644  True    True    True      True    True    True    True      True
0x000000001e2c7060  svchost.exe         764   True    True    True      True    True    True    True      True
0x000000001da11b30  explorer.exe        2336  True    True    True      True    True    True    True      True
0x000000001e584b30  userinit.exe        2268  True    True    True      True    True    True    True      True
0x000000001e302710  svchost.exe         860   True    True    True      True    True    True    True      True
0x000000001de25b30  svchost.exe         1656  True    True    True      True    True    True    True      True
0x000000001e32b9e0  svchost.exe         896   True    True    True      True    True    True    True      True
0x000000001d603b30  malware.exe         3180  True    True    True      True    True    True    True      True
0x000000001d831280  wordpad.exe         3116  True    True    True      True    True    True    True      True
0x000000001df63a10  dwm.exe             2292  True    True    True      True    True    True    True      True
```

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우회 방법

5) Rootkit 탐지 우회

▶ psxview

▶ deskthrd

EPROCESS

0x170	ExitTime
0x1e0	SessionProcessLinks
0x258	Win32Process

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psxview
```

```
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001de37580	VGAAuthService.	1688	True	True	True	True	True	True	True	
0x000000001e5f7b30	lsass.exe	512	True	True	True	True	True	True	False	
0x000000001da1bb30	vmtoolsd.exe	2644	True	True	True	True	True	True	True	
0x000000001e2c7060	svchost.exe	764	True	True	True	True	True	True	True	
0x000000001da11b30	explorer.exe	2336	True	True	True	True	True	True	True	
0x000000001e584b30	userinit.exe	2268	True	True	True	True	True	True	True	
0x000000001e302710	svchost.exe	860	True	True	True	True	True	True	True	
0x000000001de25b30	svchost.exe	1656	True	True	True	True	True	True	True	
0x000000001e32b9e0	svchost.exe	896	True	True	True	True	True	True	True	
0x000000001d603b30	malware.exe	3180	True	True	True	True	True	True	False	
0x000000001d831280	wordpad.exe	3116	True	True	True	True	True	True	True	
0x000000001df63a10	dwm.exe	2292	True	True	True	True	True	True	True	

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우회 방법

5) Rootkit 탐지 우회

▶ psxview

```

C:\volatility-master>vol.py -f C:\test.vmem --profile=Win7SP0x64 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)      Name                PID  pslist  psscan  thrddproc  pspcid  csrss  session  deskthrd  ExitTime
-----
0x000000001de37580  VGAuthService.      1688  True    True    True       True    True   True     True      True
0x000000001e5f7b30  lsass.exe           512   True    True    True       True    True   True     False     True
0x000000001da1bb30  vmtoolsd.exe        2644  True    True    True       True    True   True     True      True
0x000000001e2c7060  svchost.exe         764   True    True    True       True    True   True     True      True
0x000000001da11b30  explorer.exe        2336  True    True    True       True    True   True     True      True
0x000000001e584b30  userinit.exe        2268  True    True    True       True    True   True     True      True
0x000000001e302710  svchost.exe         860   True    True    True       True    True   True     True      True
0x000000001de25b30  svchost.exe         1656  True    True    True       True    True   True     True      True
0x000000001e32b9e0  svchost.exe         896   True    True    True       True    True   True     True      True
0x000000001d603b30  malware.exe         3180  True    True    True       True    True   True     True      True
0x000000001d831280  wordpad.exe         3116  True    True    True       True    True   True     True      True

```

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우회 방법

5) Rootkit 탐지 우회

▶ psxview

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP0x64 psxview
Volatility Foundation Volatility Framework 2.6
```

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001de37580	VGAAuthService.	1688	True	True	True	True	True	True	True	
0x000000001e5f7b30	lsass.exe	512	True	True	True	True	True	True	False	
0x000000001da1bb30	vmtoolsd.exe	2644	True	True	True	True	True	True	True	
0x000000001e2c7060	svchost.exe	764	True	True	True	True	True	True	True	
0x000000001da11b30	explorer.exe	2336	True	True	True	True	True	True	True	
0x000000001e584b30	userinit.exe	2268	True	True	True	True	True	True	True	
0x000000001e302710	svchost.exe	860	True	True	True	True	True	True	True	
0x000000001de25b30	svchost.exe	1656	True	True	True	True	True	True	True	
0x000000001e32b9e0	svchost.exe	896	True	True	True	True	True	True	True	
0x000000001d603b30	malware.exe	3180	True	True	True	True	True	True	True	
0x000000001d831280	wordpad.exe	3116	True	True	True	True	True	True	True	

```
C:\#volatility-master>vol.py -f C:\#test.vmem --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
```

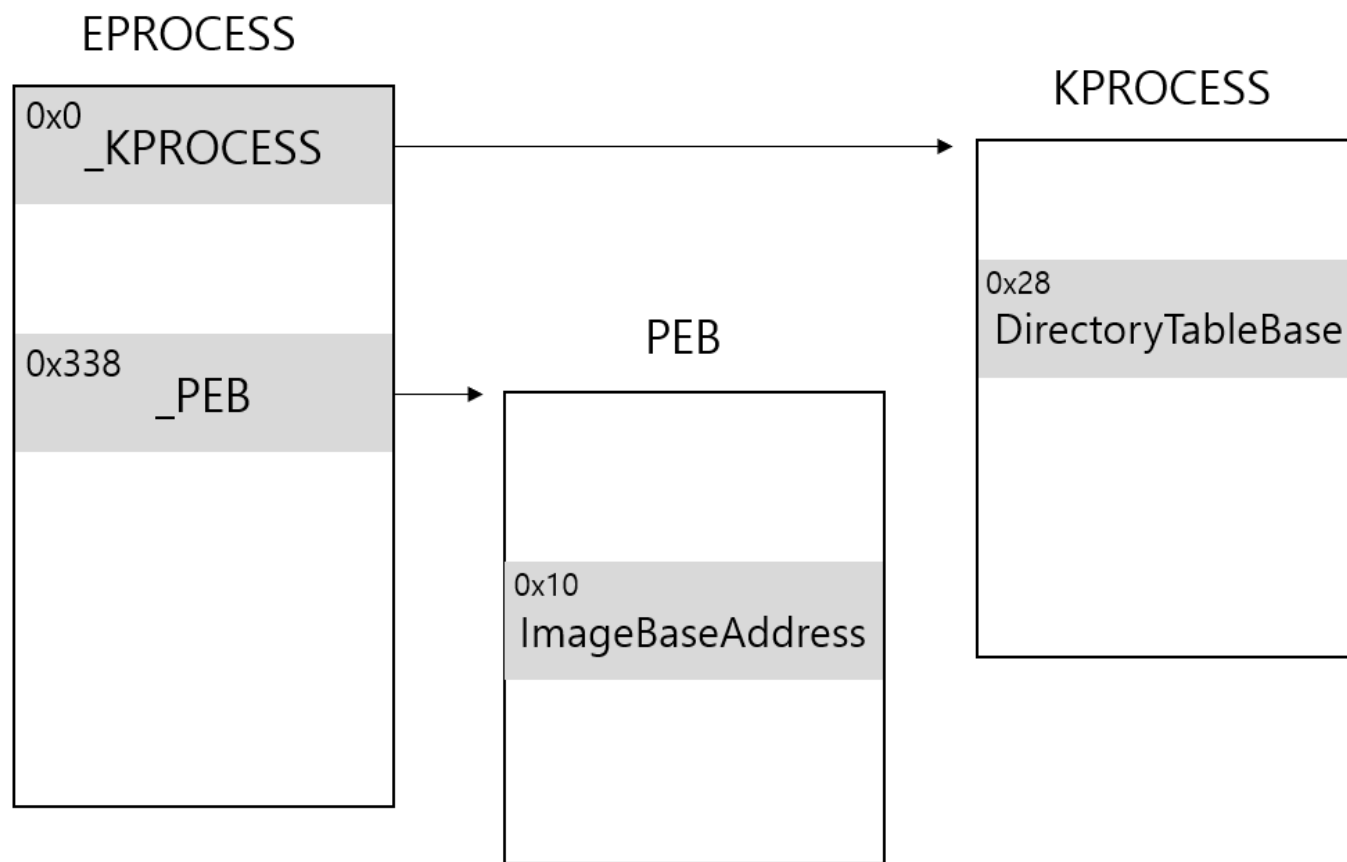
Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x000000001de37580	VGAAuthService.	1688	True	True	True	True	True	True	True	
0x000000001e5f7b30	lsass.exe	512	True	True	True	True	True	True	False	
0x000000001da1bb30	vmtoolsd.exe	2644	True	True	True	True	True	True	True	
0x000000001e2c7060	svchost.exe	764	True	True	True	True	True	True	True	
0x000000001da11b30	explorer.exe	2336	True	True	True	True	True	True	True	
0x000000001e584b30	userinit.exe	2268	True	True	True	True	True	True	True	
0x000000001e302710	svchost.exe	860	True	True	True	True	True	True	True	
0x000000001de25b30	svchost.exe	1656	True	True	True	True	True	True	True	
0x000000001e32b9e0	svchost.exe	896	True	True	True	True	True	True	True	
0x000000001d831280	wordpad.exe	3116	True	True	True	True	True	True	True	

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우회 방법

6) 프로세스 수집 우회

▶ procdump



3

우회 방법

6) 프로세스 수집 우회

▶ procdump

```
C:\volatility-master>vol.py -f C:\test.vmem --profile=Win7SP0x64 procdump -p 3180 -D C:\proc
Volatility Foundation Volatility Framework 2.6
Process(V)      ImageBase      Name      Result
-----
0xffffffffa8009403b30 0x00000000ff830000 malware.exe OK: executable.3180.exe
```

<조작 전>

```
C:\volatility-master>vol.py -f C:\test.vmem --profile=Win7SP0x64 procdump -p 3180 -D C:\proc
Volatility Foundation Volatility Framework 2.6
Process(V)      ImageBase      Name      Result
-----
0xffffffffa8009403b30 0x00000000ff4f0000 malware.exe OK: executable.3180.exe
```

<조작 후>

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우회 방법

6) 프로세스 수집 우회

▶ procdump

```
C:\wvolatility-master>vol.py -f C:\wtest.vmem --profile=Win7SP0x64 procdump -p 3180 -D C:\wproc
Volatility Foundation Volatility Framework 2.6
Process(V)      ImageBase      Name      Result
-----
0xffffffffa8009403b30 0x00000000ff830000 malware.exe OK: executable.3180.exe
```

<조작 전>

```
C:\wvolatility-master>vol.py -f C:\wtest.vmem --profile=Win7SP0x64 procdump -p 3180 -D C:\wproc
Volatility Foundation Volatility Framework 2.6
Process(V)      ImageBase      Name      Result
-----
0xffffffffa8009403b30 0x00000000ff4f0000 malware.exe OK: executable.3180.exe
```

<조작 후>

```
C:\wvolatility-master>vol.py -f C:\wtest.vmem --profile=Win7SP0x64 procdump -p 3116 -D C:\wproc
Volatility Foundation Volatility Framework 2.6
Process(V)      ImageBase      Name      Result
-----
0xffffffffa8009231280 0x00000000ff4f0000 wordpad.exe OK: executable.3116.exe
```

<정상 프로세스>

<35/43>

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우회 방법

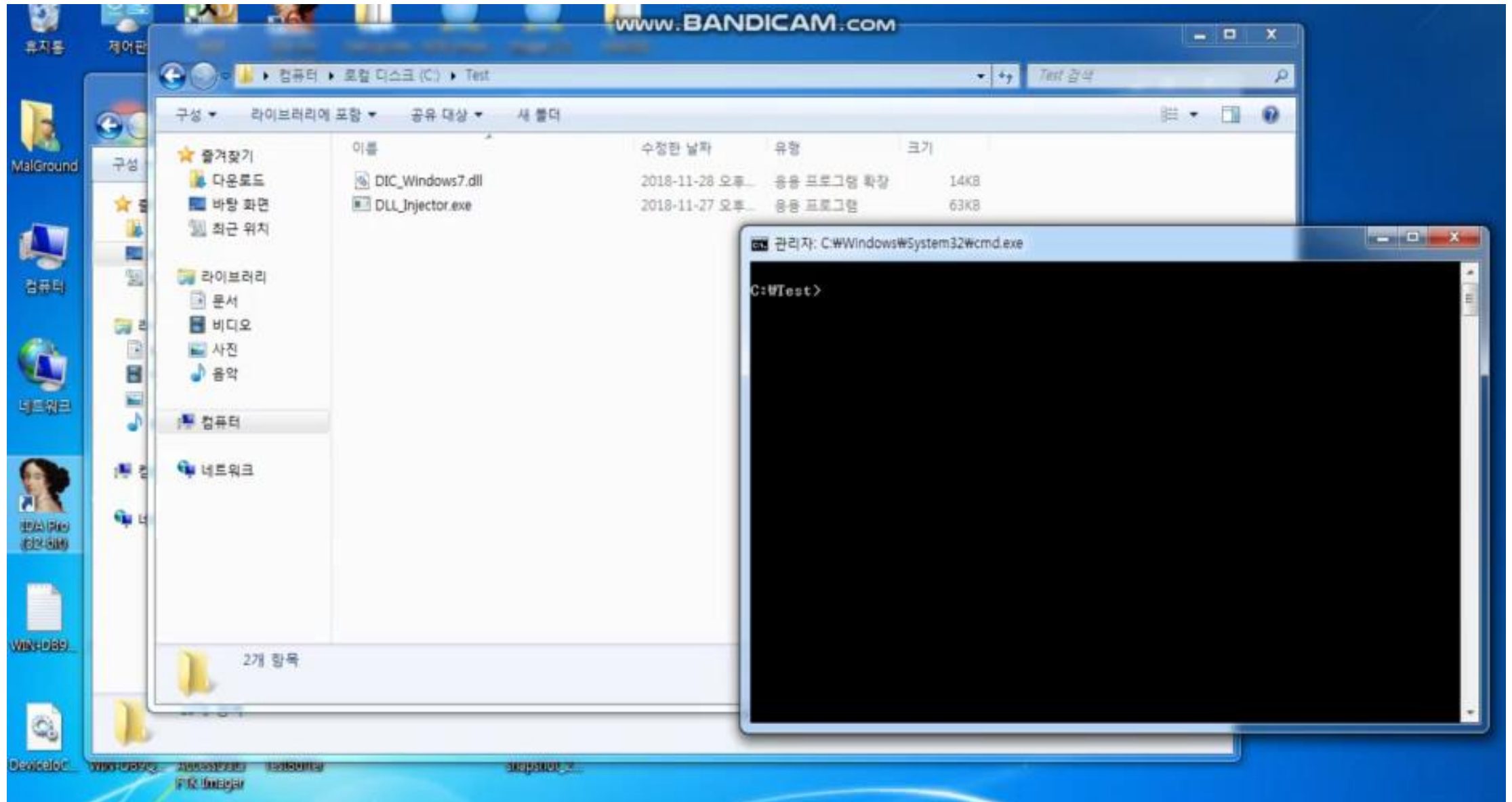
6) 프로세스 수집 우회

 procdump

[해시 값 비교]

파일	종류	해시 값
① malware.exe	SHA-256	0573871D81A09F4C92AE5F28BE062866BA92E9DC09F8E1A9A3 C193AB1D36BBE5
② wordpad.exe		0573871D81A09F4C92AE5F28BE062866BA92E9DC09F8E1A9A3 C193AB1D36BBE5

 malware.exe 대신 wordpad.exe가 추출됨



	명령어	우회 방법	Volatility	Rekall
프로세스와 DLL	pstree	EPROCESS 구조체의 InheritedFromUniqueProcessId 필드 변조	◎	◎
	psscan	POOL_HEADER : Proã (0xE36F7250) 변조	◎	◎
	dlllist	1) PEB_LDR_DATA 구조체 주소 변조 2) LDR_DATA_TABLE_ENTRY의 InLoadOrderLinks, InMemoryOrderLinks, InInitializationOrderLinks 변조	◎	-
프로세스 메모리	procdump	PEB 구조체의 ImageBaseAddress 필드 & DTB 변조	◎	◎

◎ : 명령어 우회 성공(우회 대상이 해당 도구에서 탐지되지 않음)

- : 명령어 우회 실패(우회 대상이 해당 도구에서 탐지됨)

	명령어	우회 방법	Volatility	Rekall
커널 메모리와 오브젝트	modules	KDDEBUGGER_DATA64 구조체의 PsLoadedModuleList 필드 변조	◎	-
	filescan	FILE_OBJECT : Filå (0xE56C6946) 변조	◎	◎
	driverscan	DRIVER_OBJECT : Driö (0xF6697244) 변조	◎	◎
네트워크	netscan	1) TCP_LISTENER : TcpL (0x5463704C) 변조 2) TCP_ENDPOINT : TcpE (0x54637045) 변조 3) UDP_ENDPOINT : UdpA (0x55647041) 변조	◎	◎

	명령어	우회 방법	Volatility	Rekall
레지스트리	hivescan	1) CMHIVE : CM10 (0x434D3130) 변조 2) 0xE0BEE0BE 변조	◎	◎
악성코드, 루트킷	malfind	1) MMADDRES_NODE 구조체 필드 변조 2) Tag 필드 값이 "VadS"이며 페이지에 쓰기, 실행 권한이 있을 경우, "Vad" 또는 "VadI"로 변조	◎	◎
	apihooks	LDR_DATA_TABLE_ENTRY 시작 주소 변조	◎	◎
	psxview	1) OBJECT_HEADER 구조체의 변수 조작(TypeIndex) 2) EPROCESS 구조체의 변수 조작 (ExitTime,ActiveProcessLinks,SessionProcessLinks,Win32Process)	◎	-

3장에서 소개한 명령어 6개 중,

psscan

netscan

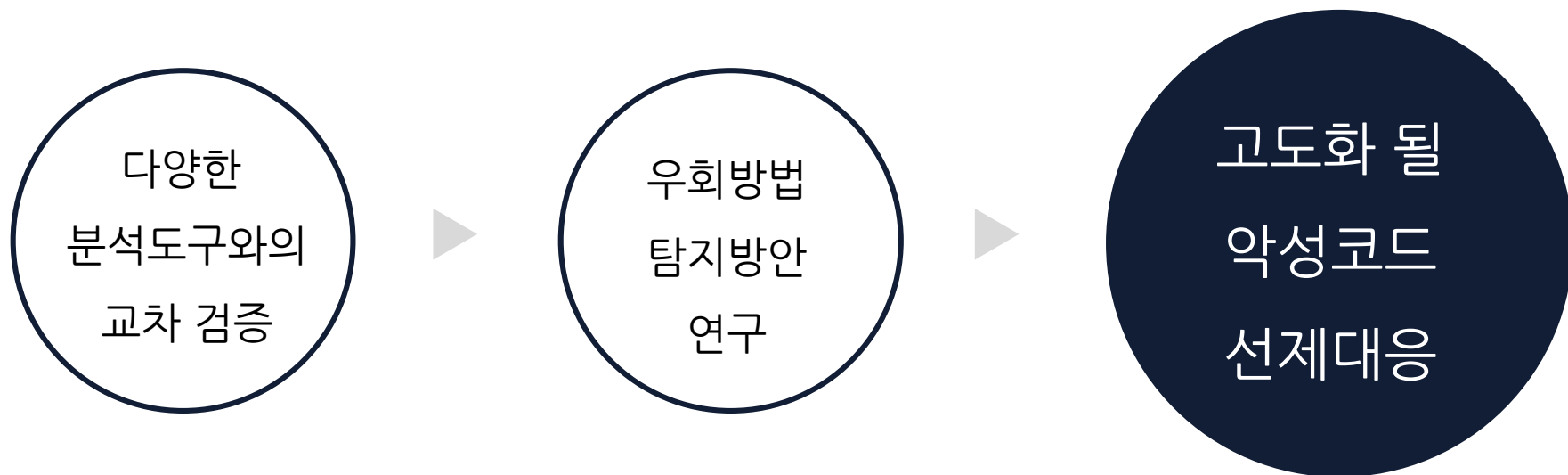
malfind

procdump

명령어 4개는 Volatility와 Rekall에서 모두 우회되었다.

메모리 덤프 파일의 조작에 따라
메모리 분석도구의 결과가 변조되거나 은폐될 수 있다.

따라서 교차 검증이 필요하며,
조작 여부를 판단할 수 있는 방안에 대한 연구가 필요하다.





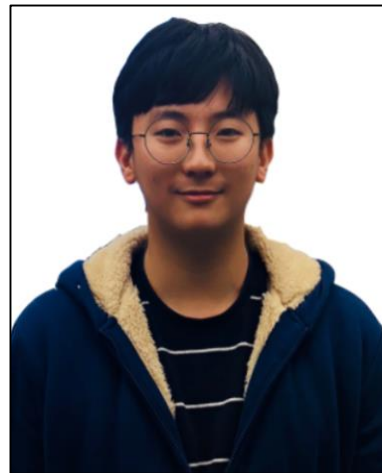
이 한 열 (Haneol Lee)



전 은 진 (Eunjin Jeon)



김 태 우 (Taewoo Kim)



김 한 솔 (Hansol Kim)



한 지 혁 (Jihyeok Han)



김 종 민 (Jongmin Kim)



김 종 현 (Jonghyun Kim)

감사합니다
